

Exam format and data checks

Multiple choice risk

Score in observed exams: correct = 3, wrong = -1, don't know = 0. Guess only if $P(\text{correct}) > 1/4$. One false clause makes a statement false. Eliminate by units, sign, impossible range, wrong dimension, and violated assumptions before computing.

Data matrix and standardization

Use $X \in \mathbb{R}^{N \times M}$: rows observations, columns attributes. Standardized attribute:

$$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}, \quad \bar{x}_j = \frac{1}{N} \sum_i x_{ij}, \quad s_j^2 = \frac{1}{N-1} \sum_i (x_{ij} - \bar{x}_j)^2.$$

Trap: If a problem says standardized, covariance and correlation coincide; diagonal variance is 1. Never standardize labels y unless explicitly stated.

Covariance, correlation, distance

$$\hat{\Sigma} = \frac{1}{N-1} \tilde{X}^\top \tilde{X}, \quad \rho_{jk} = \frac{\text{Cov}(x_j, x_k)}{s_j s_k},$$

$$d(x_i, x_j) = \sqrt{\sum_{m=1}^M (x_{im} - x_{jm})^2}.$$

Check: Correlation matrix is symmetric, diagonal 1, entries in $[-1, 1]$. Euclidean distance is nonnegative, symmetric, diagonal 0.

Visual/statistical identification

Question cue	Fast check
Histogram vs percentile	Match support, skew, spikes, median and tails; standardized values can still be non-Gaussian.
Boxplot	Median line, IQR box, whiskers/outliers; compare sign after standardization.
Scatter matrix	Sign from slope/cloud tilt; strength from narrowness; density color not class unless stated.
Candidate matrix	Symmetry + diagonal first, then signs from plots, then magnitudes.

Losses and simple estimators

$$\text{MSE} = \frac{1}{N} \sum_i (y_i - \hat{y}_i)^2, \quad \text{RMSE} = \sqrt{\text{MSE}}, \quad \text{SSE} = \sum_i e_i^2.$$

Trap: Training error normally decreases with model flexibility; test error can increase. For constant regression predictor use mean of training y , not test y .

PCA and linear algebra

SVD/PCA notation

For centered or standardized $\tilde{X}$:

$$\tilde{X} = USV^\top, \quad V = [v_1, \dots, v_M], \quad Z = \tilde{X}V = US.$$

Projection of observation $\tilde{x}_i$ on first K PCs:

$$z_i = \tilde{x}_i V_K, \quad \hat{x}_i = z_i V_K^\top + \bar{x}.$$

Trap: Exam matrices often print V ; PC k is column k , not row k . Sign of a PC is arbitrary, but projections and loadings must use the same sign.

Explained variance

$$\lambda_k = \frac{s_k^2}{N-1}, \quad \text{EVR}_k = \frac{s_k^2}{\sum_j s_j^2}, \quad \text{cumEVR}(K) = \frac{\sum_{k=1}^K s_k^2}{\sum_j s_j^2}.$$

Use: Choose fewest K with cumulative explained variance above target (often 90%). **Check:** If standardized, total variance $\approx M$ and $\|\tilde{X}\|_F^2 = \sum_k s_k^2 \approx (N-1)M$.

Reading PC directions

$$z_{ik} = \tilde{x}_i^\top v_k = \sum_m \tilde{x}_{im} v_{mk}.$$

Recipe: To evaluate a statement: identify signs/magnitudes of $\tilde{x}_m$, multiply by column v_k , sum dominant terms, compare sign. **Trap:** "High original value" means positive standardized value only if above training mean.

PCA plot candidates

Object	Rule
Scores Z	Points in PC coordinate system.
Loading vector v_k	Attribute direction; positive attributes increase with arrow direction.
Biplot sign	Flipping all signs of one PC gives equivalent plot.
Orthogonality	PC directions are orthogonal in attribute space.
Reconstruction	More PCs never increases reconstruction error.

Common linear algebra checks

$$\|x\|_2 = \sqrt{x^\top x}, \quad \|X\|_F^2 = \sum_{ij} x_{ij}^2 = \sum_k s_k^2, \quad \cos(x, y) = \frac{x^\top y}{\|x\| \|y\|}.$$

Trap: Cosine ignores magnitude; Euclidean distance does not. Sparse text vectors often compare with cosine or Jaccard, not raw Euclidean distance.

Probability, Bayes and Naive Bayes

Probability rules

$$P(A|B) = \frac{P(A, B)}{P(B)}, \quad P(A, B) = P(A|B)P(B),$$

$$P(A) = \sum_c P(A|c)P(c).$$

$$P(y = c|x) = \frac{P(x|y = c)P(y = c)}{\sum_{c'} P(x|y = c')P(y = c')}.$$

Trap: Normalize over classes after computing unnormalized scores. Probabilities must sum to 1.

Bayes classifier from counts

$$\hat{P}(y = c|x = a) = \frac{\#\{i : y_i = c, x_i = a\}}{\#\{i : x_i = a\}}, \quad \hat{y} = \arg \max_c \hat{P}(y = c|x).$$

Use: For exam tables with binary attributes and no independence assumption. **Trap:** Condition on exactly the attributes named in the question.

Naive Bayes

$$\hat{y} = \arg \max_c P(y = c) \prod_{m=1}^M P(x_m|y = c).$$

For binary attributes:

$$\hat{P}(x_m = a|y = c) = \frac{\#\{i : x_{im} = a, y_i = c\}}{\#\{i : y_i = c\}}.$$

Condition: Assumes conditional independence of attributes given class. **Trap:** If any factor is zero, the product becomes zero unless smoothing is specified.

Gaussian class density

p(x|c) = 1 / ((2*pi)^{M/2} |Sigma_c|^{1/2}) * exp[-1/2 * (x - mu_c)^T Sigma_c^{-1} * (x - mu_c)]

Decision: Use largest p(x|c)P(c); equal priors reduce to largest density. Check: In 1D, larger variance gives wider/lower peak; far from mean gives smaller density.

Log and numerical stability

log P(y = c|x) proportional to log P(y = c) + sum_m log P(x_m|y = c).

Use: Products of many probabilities are easier to compare as log-sums.

Distributions recurring in density questions

N(x|mu, sigma^2) = 1 / (sqrt(2*pi)*sigma) * exp[-(x - mu)^2 / (2*sigma^2)]

Trap: Kernel width sigma vs variance sigma^2; exams may state "variance sigma^2".

Regression and classification models

Linear and ridge regression

Augment with intercept: x-tilde = [1, x_1, ..., x_M].

y-hat = x-tilde^T w, w_LS = (X^T X)^{-1} X^T y, w_lambda = (X^T X + lambda I)^{-1} X^T y.

Trap: Usually do not regularize intercept. Larger lambda shrinks weights; predictions move toward intercept/mean. Check: Weight vector length must match augmented input dimension.

Regularization and feature selection

Method	Exam rule
Ridge	Penalizes sum_{j>0} w_j^2; keeps all features but shrinks.
Forward selection	Start empty, add feature giving best validation error.
Backward selection	Start full, remove feature giving best validation error.
Polynomial/log features	Still linear in weights, nonlinear in original inputs.

Trap: Selection must be inside validation loop when estimating generalization.

Logistic regression

p-hat = P(y = 1|x) = sigma(x-tilde^T w) = 1 / (1 + exp(-x-tilde^T w)), y-hat = floor(p-hat >= tau).

log(p-hat / (1 - p-hat)) = x-tilde^T w.

Boundary: At tau = 0.5, decision boundary is x-tilde^T w = 0. Trap: Positive class depends on problem statement, e.g. y = 2 may be positive.

K-nearest neighbours

Task	Prediction
Classification	Majority vote among K nearest; ties require stated rule.
Regression	Mean of neighbours' y values.
Complexity	Small K: flexible/high variance. Large K: smoother/high bias.

Trap: Distance is computed in input space only; standardization can change neighbours.

Multiclass and one-vs-all

p-hat_c = exp(a_c) / sum_j exp(a_j), y-hat = arg max_c p-hat_c.

Check: Softmax probabilities sum to 1. Binary logistic is the two-class special case.

Model recognition from decision boundary

Boundary/behavior	Likely model
Straight line/plane	Logistic or linear classifier.
Axis-aligned steps	Decision tree.
Voronoi-like jagged regions	KNN, especially small K.
Smooth curved boundary	Feature transform, ANN, kernel/GMM.

Validation, metrics and ROC

Hold-out and K-fold

E_test = 1/|T| * sum_{i in T} L(y_i, f-hat(x_i)), E_CV = 1/K * sum_{k=1}^K E_k.

Condition: Preprocessing, feature selection, upsampling and model selection must be fit on training data only. Trap: Test set is for final evaluation, not tuning.

One-level vs two-level CV

Purpose	Correct use
Model selection	Inner/one-level CV compares hyperparameters/models.
Performance of selected procedure	Outer test folds in two-level CV.
Train count	Outer folds x inner candidates/folds plus final fits if requested.

Trap: Reporting the best inner validation error is optimistically biased.

Confusion matrix

For positive class:

Matrix with y-hat = + and y-hat = - on columns, y = + and y = - on rows. Elements are TP, FN, FP, TN.

Acc = (TP + TN) / N, Recall/TPR = TP / (TP + FN), Precision = TP / (TP + FP), FPR = FP / (FP + TN), Specificity = TN / (TN + FP), F1 = (2PR) / (P + R).

Trap: Always identify the positive class from the question. High accuracy can be useless with class imbalance.

ROC and threshold

ROC point at threshold tau = (FPR(tau), TPR(tau)).

Recipe: Sort scores high-to-low; lowering threshold adds predicted positives; update TP or FP; plot (FPR, TPR). Check: Random classifier AUC approx 0.5; perfect AUC = 1. ROC starts (0, 0) and ends (1, 1).

Comparing regression models

Statement	Check
“Always better”	Need loss no larger on every relevant test point/fold.
Mean vs median	Squared loss favors mean; absolute loss favors median.
Overfit	Low training error, high validation/test error.
Regularization choice	Pick lowest validation/CV error, not lowest training error.

Decision trees, ensembles and ANN

Decision tree splits

Classification impurity examples:

class. error = 1 - max_c p_c, Gini = 1 - sum_c p_c^2,

entropy = - sum_c p_c log_2 p_c.

gain = I(parent) - sum_v (N_v/N) I(child v).

Recipe: For a candidate split: count classes in each branch, compute weighted child impurity, choose largest gain. **Trap:** Use branch weights N_v/N; pure child impurity is 0.

Tree prediction and pruning

Recipe: Follow threshold tests from root to leaf; predict majority class or leaf mean. **Trap:** Axis-aligned splits compare one attribute at a time; depth/leaf count controls complexity.

AdaBoost

epsilon_t = sum_i w_i^{(t)} I[h_t(x_i) != y_i], alpha_t = 1/2 ln (1 - epsilon_t) / epsilon_t,

w_i^{(t+1)} proportional to w_i^{(t)} exp(alpha_t I[h_t(x_i) != y_i]), H(x) = sign(sum_t alpha_t h_t(x)).

Trap: Misclassified observations get larger weight. Need epsilon_t < 0.5 for positive alpha_t.

Neural network forward pass

Two-layer regression/classification template:

h_j = g(w_{0j}^{(1)} + sum_m w_{mj}^{(1)} x_m), y_hat = w_0^{(2)} + sum_j w_j^{(2)} h_j.

ReLU(a) = max(0, a), tanh(a) in [-1, 1],
sigma(a) in (0, 1).

Recipe: Compute hidden pre-activations, apply activation, then output layer. Include bias terms if listed. **Trap:** For classification, final activation/threshold may differ from regression output.

ANN and regularization checks

Cue	Rule
More hidden units/layers	More flexible; more overfit risk.
Weight decay/regularization	Shrinks weights; smoother boundary.
Nonlinear activation absent	Network collapses to linear model.

Model statement traps

Common false statements
Regularization does not make a linear-in-parameters model nonlinear. K-means/GMM are unsupervised; logistic regression/ANN/trees are supervised when trained with labels. A method can have lower test error without being uniformly better on every observation.

Clustering and density/outliers

K-means

min_{C_k, mu_k} sum_{k=1}^K sum_{i in C_k} ||x_i - mu_k||^2, mu_k = 1/|C_k| sum_{i in C_k} x_i.

Recipe: Assign to nearest centroid, recompute centroids, repeat until unchanged. **Trap:** Sensitive to initialization and scaling; finds center-based/spherical clusters.

Hierarchical clustering

Linkage	Distance between clusters
Single	min d(x_i, x_j); chaining/contiguity.
Complete	max d(x_i, x_j); compact clusters.
Average	Average pairwise distance.
Centroid	Distance between centroids.

Recipe: Merge closest clusters step by step; dendrogram height is merge distance. Cut horizontal line to get clusters.

Clustering comparison

Pair counts: f_11 same in both, f_00 different in both, f_10, f_01 disagreements, K = N(N - 1)/2.

Rand = (f_11 + f_00) / K, Jaccard = f_11 / (f_11 + f_10 + f_01).

Trap: Rand includes agreements on different-cluster pairs; Jaccard ignores f_00.

Gaussian mixture model

p(x) = sum_{k=1}^K pi_k N(x | mu_k, Sigma_k), sum_k pi_k = 1,
gamma_ik = (pi_k N(x_i | mu_k, Sigma_k)) / (sum_j pi_j N(x_i | mu_j, Sigma_j)).

Decision: Assign to component with largest responsibility gamma_ik. **Trap:** Mixture density is sum of weighted component densities, not max.

Kernel density estimation

p_hat(x) = 1/N sum_{i=1}^N N(x | x_i, sigma^2 I).

Leave-one-out density for selecting sigma:

L(sigma) = sum_i log (1/(N-1) sum_{j != i} N(x_i | x_j, sigma^2 I)).

Outlier: Potential outlier has low estimated density. **Trap:** Larger sigma smooths; smaller sigma gives spiky density.

KNN density and ARD

Let N_{X \setminus i}(x_i, K) be the K nearest neighbours of x_i excluding itself.

density(x_i, K) = (1/K sum_{x_j in N_i} d(x_i, x_j))^{-1},

$$\text{ARD}(x_i, K) = \frac{\text{density}(x_i, K)}{\frac{1}{K} \sum_{x_j \in N_i} \text{density}(x_j, K)}.$$

Outlier: Low density or low ARD. ARD compares to local neighbourhood density. **Trap:** Recompute each neighbour's density excluding that neighbour itself.

When in doubt

Check dimension first, then sign, then range. In 02450-style options, many wrong answers are dimension mismatches, missing intercepts, using rows instead of columns of V , forgetting to normalize Bayes probabilities, or using test data during model selection.

Association mining, text and final checks

Association rules

For itemset X and rule $X \rightarrow Y$:

$$\text{supp}(X) = \frac{\#(X)}{N}, \quad \text{conf}(X \rightarrow Y) = \frac{\text{supp}(X \cup Y)}{\text{supp}(X)},$$

$$\text{lift}(X \rightarrow Y) = \frac{\text{conf}(X \rightarrow Y)}{\text{supp}(Y)}.$$

Condition: X and Y disjoint. High confidence can be uninteresting if Y is common; use lift. **Check:** Confidence in $[0, 1]$; support of larger itemset cannot exceed support of subset.

Apriori

Recipe: Generate candidates of size k from frequent $(k-1)$ -itemsets; prune any candidate with an infrequent subset; count support; keep those above minimum support. **Trap:** If an itemset is infrequent, every superset is infrequent.

Binary similarity

	$b = 1$	$b = 0$
$a = 1$	f_{11}	f_{10}
$a = 0$	f_{01}	f_{00}

$$J(a, b) = \frac{f_{11}}{f_{11} + f_{10} + f_{01}}, \quad \text{SMC} = \frac{f_{11} + f_{00}}{M}.$$

Trap: Jaccard ignores shared zeros; useful for sparse baskets/text.

Text vectors

$$\text{tf-idf}_{ij} = \text{tf}_{ij} \log \frac{N}{\text{df}_j}, \quad \cos(x, y) = \frac{x^\top y}{\|x\| \|y\|}.$$

Check: Word absent in both documents should not create Jaccard evidence.

Fast answer plausibility checklist

Object	Must be true
Probability	In $[0, 1]$; class probabilities sum to 1.
Correlation	Symmetric matrix, diagonal 1, values in $[-1, 1]$.
Distance	Nonnegative, diagonal 0, symmetric.
PCA variance	Ratios nonnegative and sum to 1.
Confusion metric	Denominator matches row/column meaning.
KDE/GMM density	Nonnegative; can exceed 1 in continuous 1D.
Regularization	Larger λ does not increase weight norm.

Recurring exam recipes

1. **Match plot to statistic:** remove impossible matrices by symmetry/range, then match signs and only then magnitudes.
2. **Compute classifier probability:** write numerator for each class, normalize, compare.
3. **Choose hyperparameter:** use validation/CV loss or LOO log density, not training loss.
4. **Read boundary plot:** identify linear/axis-aligned/jagged/smooth before reading labels.
5. **Outlier question:** compute density/ARD; smallest density or ARD is most suspicious.